Jin Hyung Lee

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Education

Purdue University, West Lafayette, IN, U.S.A.
Postdoctoral Researcher — Oct 2025 – Present
Advisor: Dr. Anindya Bhadra

George Mason University, Fairfax, VA, U.S.A.
Ph.D. in Statistics — Dec 2025
Dissertation: *Variational Approximations for Massive Spatial Data: Scalable Bayesian Inference*
Advisor: Dr. Ben Seiyon Lee

Seoul National University, Seoul, South Korea
M.P.H. in Biostatistics — Aug 2020
Advisor: Dr. Ho Kim

Yonsei University, Seoul, South Korea
B.A. in Applied Statistics & Economics — Aug 2015

Research Interests

Variational inference, Normalizing flows, Neural networks, Bayesian statistics, Spatial statistics, functional data analysis, with applications in atmospheric science, fMRI, and clinical trials.

Publications

1. **Lee, J.H.**, and Lee, B.S. (2026). A scalable variational Bayes approach for fitting non-conjugate spatial generalized linear mixed models via basis expansions. *Computational Statistics & Data Analysis*. To appear
2. **Lee, J.H.**, and Lee, B.S. (2026). A scalable variational Bayes approach to fit high-dimensional spatial generalized linear mixed models. *Technometrics*, 68(1), 146–158.
3. Poythress, J.C., **Lee, J.H.**, Takeda, K., and Liu, J. (2024). Bayesian methods for quality tolerance limit (QTL) monitoring. *Pharmaceutical Statistics*, 23(6), 1166–1180.

Under Review

1. Wang, Y., **Lee, J.H.**, Yamaguchi, Y., and Han, C. (2026). Bayesian hierarchical dose-response model averaging in small clinical trials for decision-making. *Under review.*

Manuscripts in Preparation

1. **Lee, J.H.**, and Bhadra, A. (2026). A Scalable Variational Approach for Modeling Multivariate Spatial Data with Variational Inference Normalizing Flows
2. Lee, H.S., Kim, J.S., Park, J., Lee, B.S., and **Lee, J.H.** (2026). Deep spatial function-on-scalar regression with robust nonparametric testing.
3. **Lee, J.H.**, Lee, S., and Kim, S.D. (2026). A scalable variational Bayes method for a robust high-dimensional Bayesian spatial quantile regression.

4. Hong, S., **Lee, J.H.** (2026) Dynamic Stochastic General Equilibrium Models with Variational Inference Normalizing Flows

Honors & Awards

- Winner, Korean International Statistical Society (KISS) Outstanding Student Paper Competition Jan 2025
- ENVR Travel Award to NCAR, American Statistical Association (ASA) Oct 2024
- R. Clifton Bailey Travel Award, Department of Statistics, George Mason University Aug 2024
- Graduate Student Travel Fund Award, George Mason University Aug 2024
- Graduate Student Travel Fund Award, George Mason University Aug 2023
- Distinguished Service and Leadership Award, Department of Statistics, George Mason University May 2023
- Second Place Poster Presentation Award, UMBC Probability and Statistics Day Apr 2023

Presentations

A Scalable Variational Bayes Approach for Fitting Non-Conjugate Spatial Generalized Linear Mixed Models via Basis Expansions

- Eastern North America Region (ENAR) 2026 Spring Meeting (Talk, Mar 2026). Indianapolis, IN.
- Department of Statistics Zoom Seminar at Inha University (Talk, Oct 2025). Incheon, South Korea.
- Joint Statistical Meetings (Talk, Aug 2024). Portland, OR.

A Scalable Variational Bayes Approach to Fit High-Dimensional Spatial Generalized Linear Mixed Models

- Joint Statistical Meetings (Talk, Aug 2025). Nashville, TN.
- StatConnect 2025 (Poster, Mar 2025). Fairfax, VA.
- ENVR Workshop: Spatial Data Science for the Environment, Mesa Lab at NCAR (Poster, Oct 2024). Boulder, CO.
- Joint Statistical Meetings (Talk, Aug 2024). Portland, OR.
- Joint Statistical Meetings (Talk, Aug 2023). Toronto, Canada.
- Clemson Climate Extremes Workshop (Poster, May 2023). Clemson, SC.
- UMBC 14th Probability and Statistics Day (Poster, Apr 2023). Baltimore, MD.
- 30 Years of Mason Statistics (Poster, Mar 2023). Fairfax, VA.

Bayesian Hierarchical Dose-Response Model Averaging in Small Clinical Trials for Decision Making

- 2024 NIC-ASA & ICSA Midwest Chapter Joint Fall Conference (Talk, Oct 2024). Chicago, IL.

Work Experience

Astellas Pharma US, Northbrook, IL
Biostatistics Intern May – Aug 2024

- Proposed a novel Bayesian Hierarchical Dose Response Model Averaging (BHDRMA) using Bayes factor, BIC, and Power Prior, validated via simulation against bootstrap model averaging to inform Go/No-Go decisions.

Astellas Pharma US, Northbrook, IL
Biostatistics Intern May – Aug 2023

- Proposed a novel Bayesian method based on partial information for monitoring Quality Tolerance Limit (QTL) parameters.

Korea International Trade Association, Seoul, South Korea
Senior Data Researcher Jan 2016 – Jun 2020

- Conducted in-depth analysis of global index and trade data from UN Comtrade and The World Bank by using Applied machine learning algorithms.

Academic Service

Review	Statistica Sinica, Spatial Statistics, Statistical Analysis and Data Mining	
Community	Student Representative, Korean International Statistical Society	Aug 2024
	Vice President, George Mason University Statistics Graduate Student Association	Aug 2022

Teaching

Instructor, Purdue University
STAT517 Introduction to Statistical Theory and Inference Summer 2026

Instructor, George Mason University
STAT344 Probability and Statistics for Engineers and Scientists I Summer 2025
Teaching Evaluation (4.49/5), Response Ratio 79.55%

Graduate Teaching Assistant, George Mason University

STAT350	Introductory Statistics II	2024F, 2025S
STAT572	Intro Statistical Learning	2025S
STAT346	Probability for Engineers	2024F
STAT354	Probability and Statistics for Engineers and Scientists II	2022S
STAT344	Probability and Statistics for Engineers and Scientists I	2021Su, 2022S
STAT250	Introductory Statistics I	2021F, 2022Su

Technical Skills

Programming: R, SAS, TensorFlow, PyTorch, Python, LaTeX, SQL, R Shiny, AWS, Datawrapper, Flourish, Tableau

References

Dr. Ben Seiyon Lee
Assistant Professor
Department of Statistics at George Mason University
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Dr. Anindya Bhadra
Professor
Department of Statistics at Purdue University
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Dr. Lily Wang
Professor
Department of Statistics at George Mason University
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